

DIGITAL IMAGE PROCESSING

LECTURE # 3

DIGITAL IMAGE FUNDAMENTALS-I

Contact Information

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Topics to Cover

- Sampling and Quantization
- Spatial and Gray level Resolution
- Image Scaling
 - ✓ Image Zooming
 - ✓ Image Shrinking
- Basic relationship between pixels
 - ✓ Neighbors
 - ✓ Adjacency
 - ✓ Connectivity
 - ✓ Path
 - ✓ Regions and Boundaries
 - ✓ Distance between pixels

Image Digitization

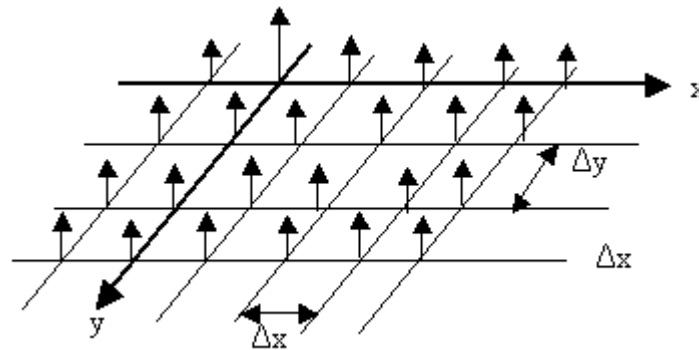
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- ❑ Digitalization of an analog signal involves two operations:
 - ✓ Sampling, and
 - ✓ Quantization.
- ❑ Both operations correspond to a discretization of a quantity, but in different domains.
- ❑ An image captured by a sensor is expressed as a continuous function $f(x,y)$ of two coordinates in the plane.
- ❑ Image digitization means that the function $f(x,y)$ is sampled into a matrix with M rows and N columns.
- ❑ The image quantization assigns to each continuous sample an integer value. The continuous range of the image function $f(x,y)$ is split into k intervals.
- ❑ The finer the sampling (i.e., the larger M and N) and quantization (the larger K) the better the approximation of the continuous image function $f(x,y)$.

Sampling

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- A continuous image function $f(x,y)$ can be sampled using a discrete grid of sampling points in the plane.
- Two neighboring sampling points are separated by distance Δx along the x axis and Δy along the y axis. Distances Δx and Δy are called the **sampling interval** and the matrix of samples constitutes the discrete image



Sampling

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- Sampling corresponds to a discretization of the space. That is, of the domain of the function, into $f : [1; :: : ;N] [1; :: : ;M]$
- Thus, the image can be seen as matrix,

$$f = \begin{bmatrix} f(1,1) & f(1,2) & \cdots & f(1,M) \\ f(2,1) & f(2,2) & \cdots & f(2,M) \\ \vdots & \vdots & \ddots & \vdots \\ f(N,1) & f(N,2) & \cdots & f(N,M) \end{bmatrix}$$

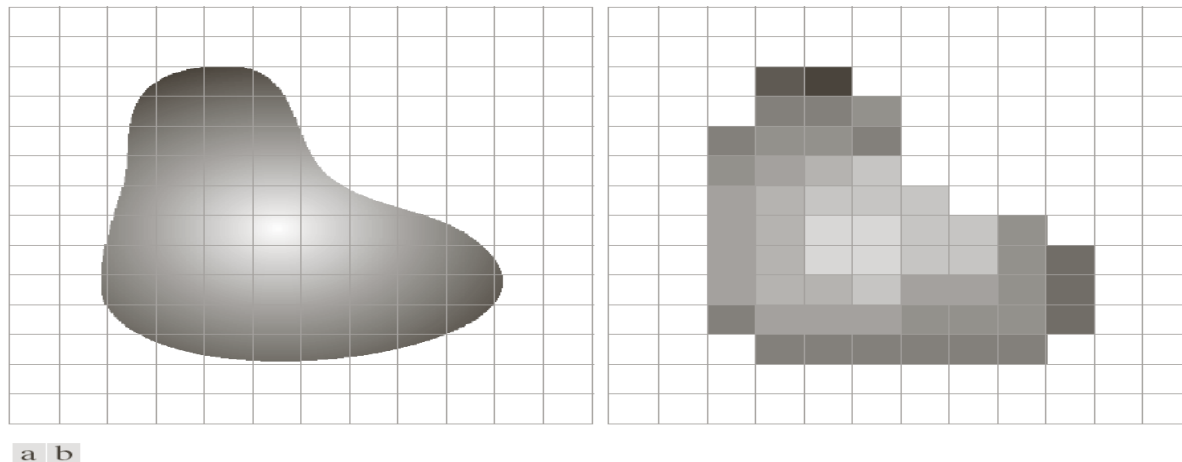


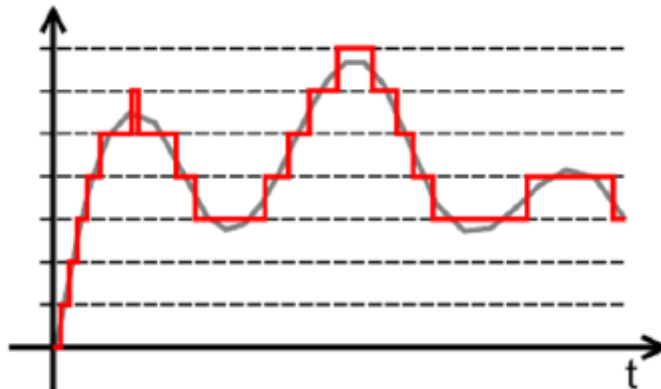
FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

Quantization

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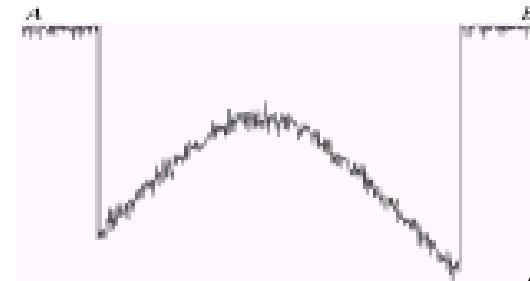
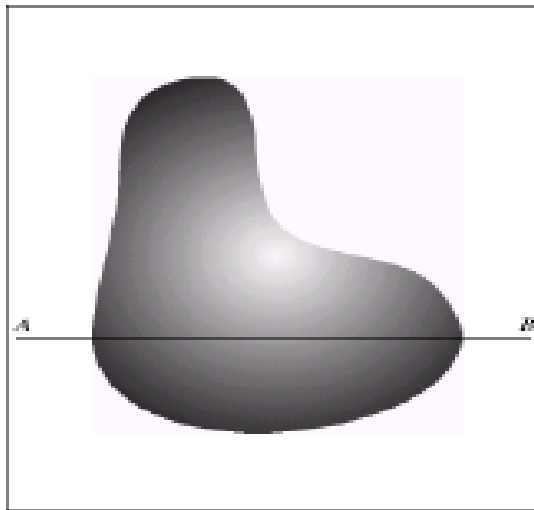
- ❑ The transition between continuous values of the image function (brightness) and its digital equivalent is called quantization.
- ❑ Quantization corresponds to a discretization of the intensity values.
- ❑ The number of quantization levels should be high enough for human perception of fine shading details in the image.
- ❑ After Sampling and Quantization, we get the following

$$f : [1; ::::; N] \ [1; ::::; M] \rightarrow [0,, L-1]$$



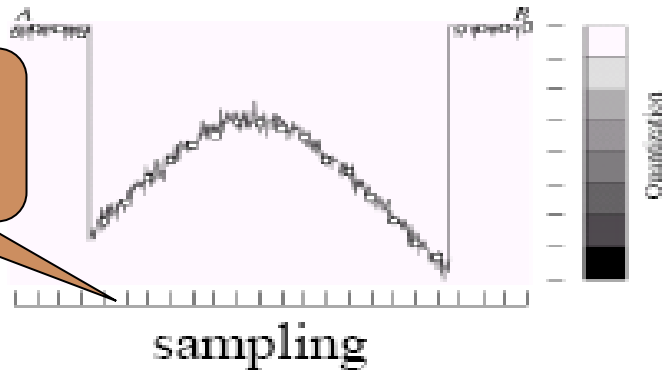
Sampling and Quantization

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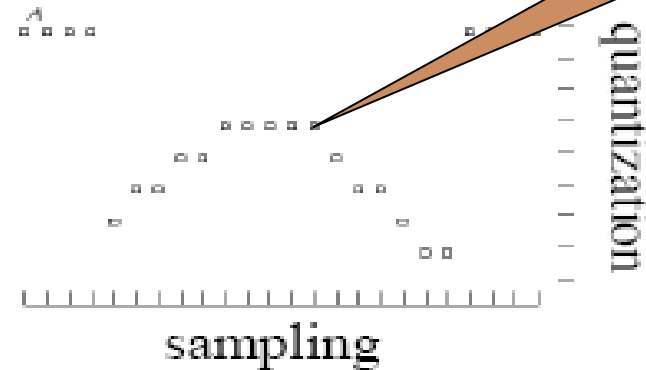


Digitizing the amplitude values

Digitizing the coordinate values



sampling



sampling

Sampling and Quantization

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- ❑ **Sampling:** Digitization of the spatial coordinates (x,y)
- ❑ **Quantization:** Digitization in amplitude (also called *gray level quantization*)
- ❑ **8 bit quantization:** $2^8 = 256$ gray levels (0: black, 255: white)
- ❑ **Binary (1 bit quantization):** 2 gray levels (0: black, 1: white)
- ❑ **Commonly used number of samples (resolution)**
- ❑ Digital still cameras: 640x480, 1024x1024, up to 4064 x 2704
- ❑ Digital video cameras: 640x480 at 30 frames/second and higher

Spatial and Gray level Resolution

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□ Spatial resolution

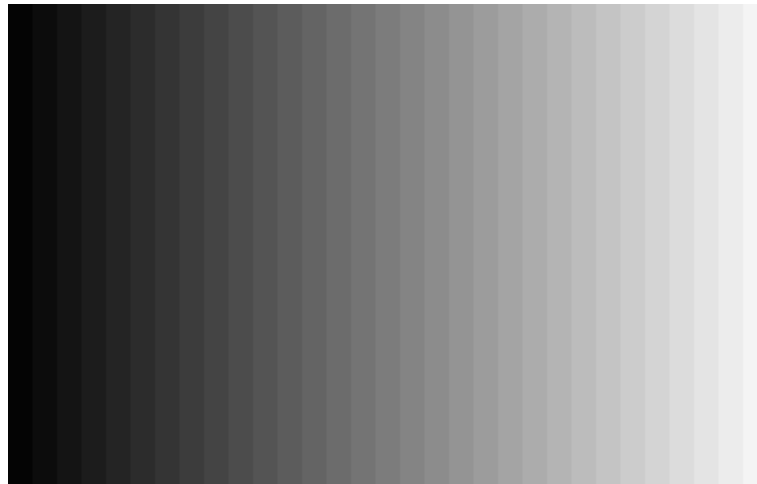
Resolution in Space

- ✓ A measure of the smallest visible detail in an image stated with *dots (pixels) per unit distance, dots per inch (dpi)*
- ✓ *No. of pixels specifies the spatial resolution*

□ Intensity or Gray level resolution

Resolution in Gray values

- ✓ The smallest visible change in intensity or gray level stated with *8 bits, 16 bits, etc.*



Spatial and Gray level Resolution

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- The Digitization process requires to determine M , N and L



- M and N are Spatial Resolution
- L = Gray level Resolution
 - $L = 2^k$, where L represents Gray level
- Image Storage = Spatial Resolution * Gray level Resolution
- The no. of bits required to store the image is::
 - $b = M \times N \times k$ or $b = N^2 \times k$
- Sampling --→ Spatial Resolution
- Quantization --→ Gray level Resolution

Spatial and Gray level Resolution

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TABLE 2.1

Number of storage bits for various values of N and k .

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

Spatial and Gray level Resolution

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Variation in Spatial Resolution from 1024 x 1024 to 32 x 32

Spatial and Gray level Resolution

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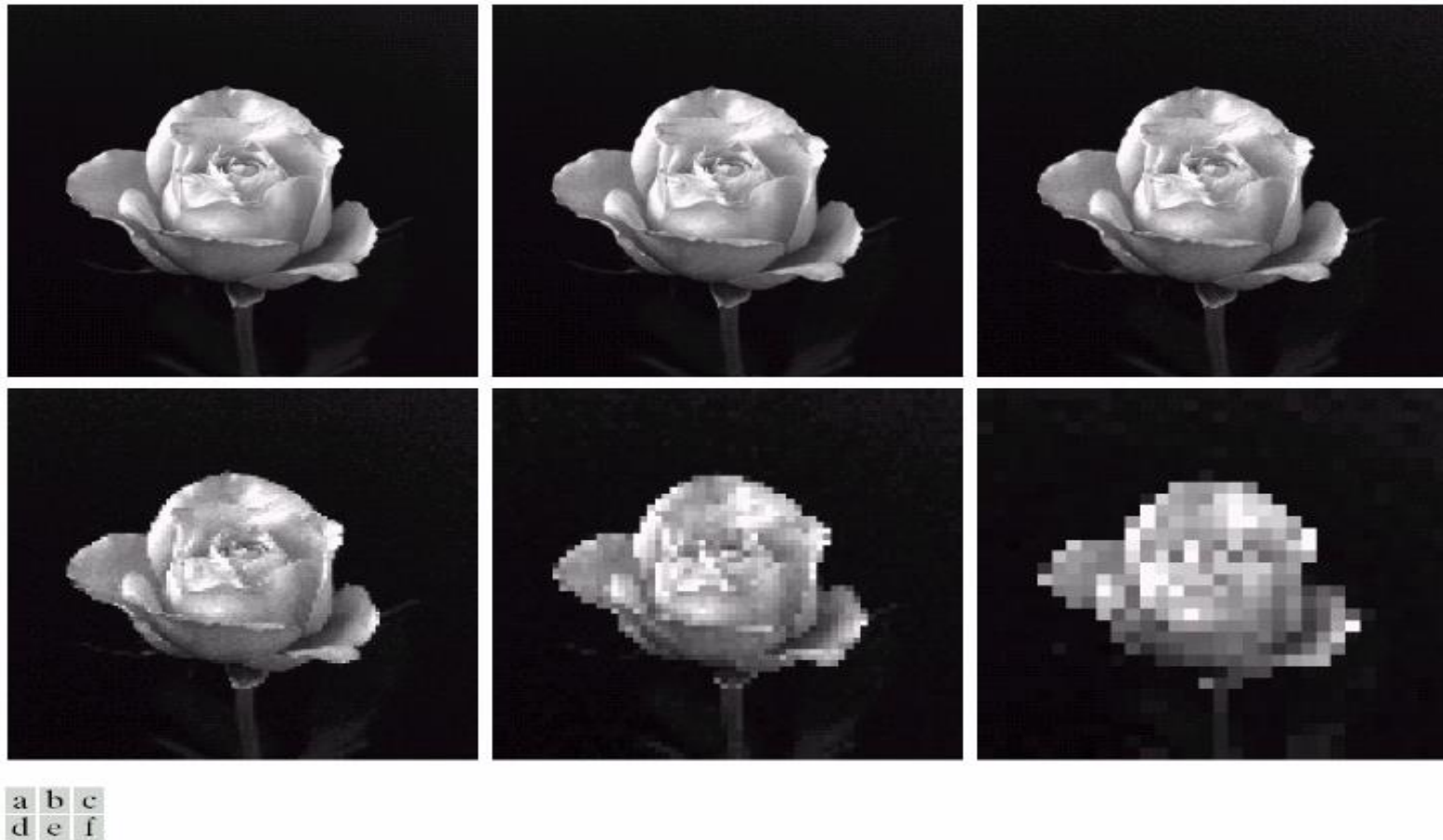
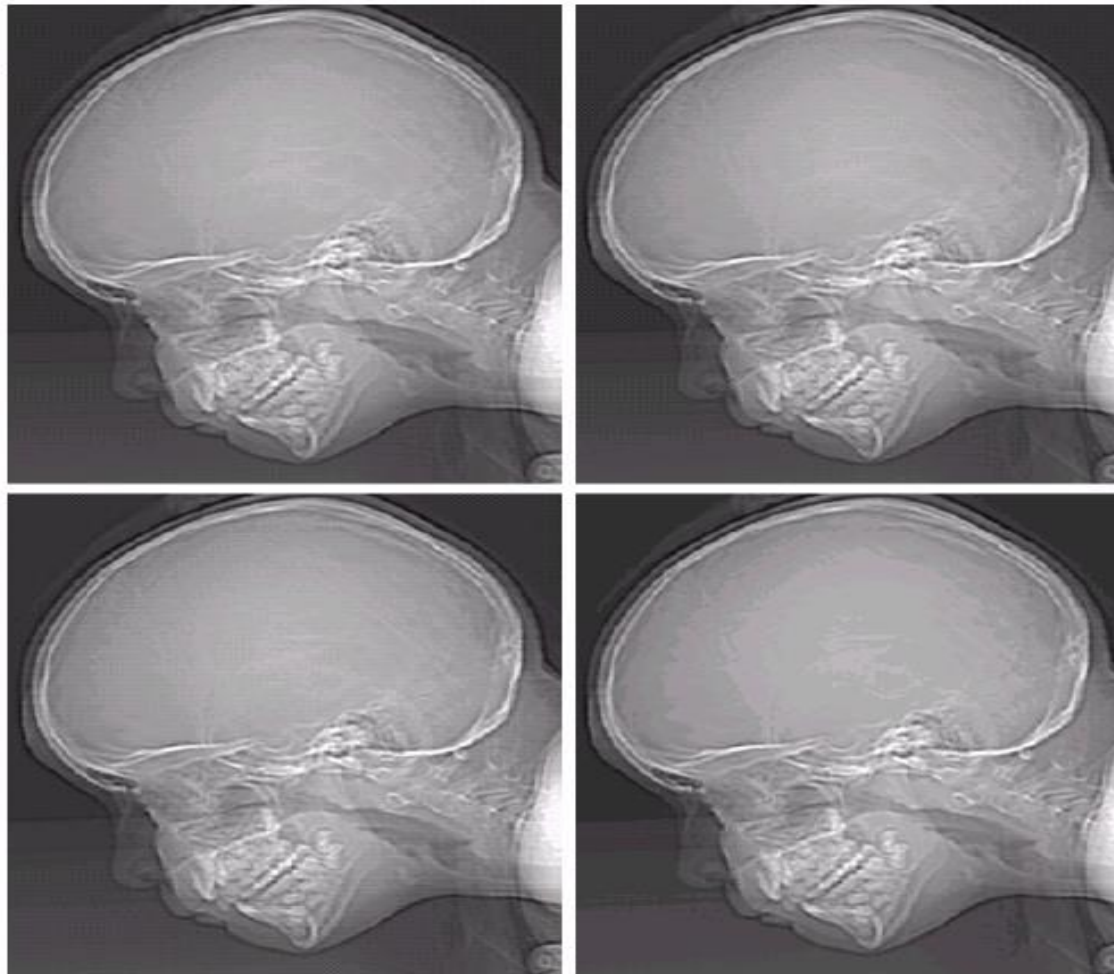


FIGURE 2.20 (a) 1024×1024 , 8-bit image. (b) 512×512 image resampled into 1024×1024 pixels by row and column duplication. (c) through (f) 256×256 , 128×128 , 64×64 , and 32×32 images resampled into 1024×1024 pixels.

Spatial and Gray level Resolution

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a b
c d

FIGURE 2.21

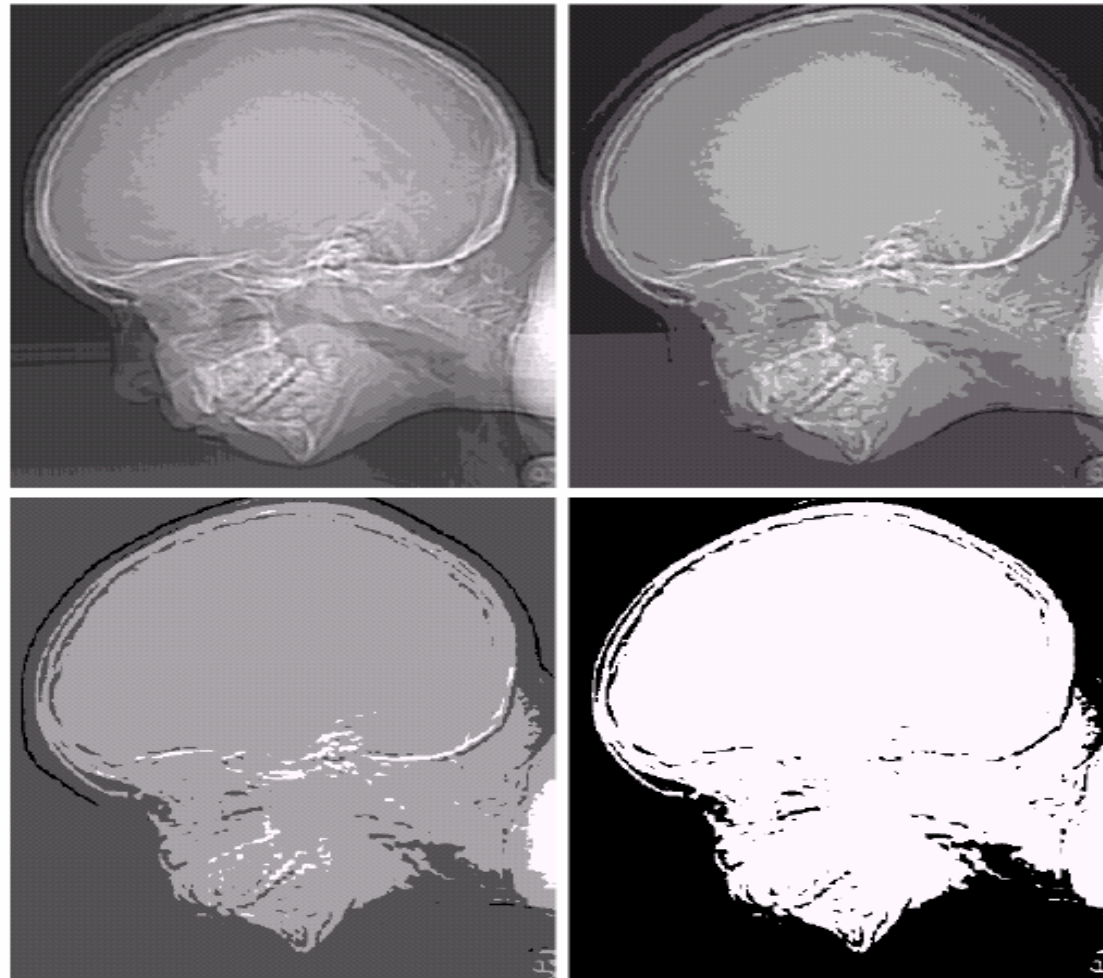
(a) 452×374 , 256-level image. (b)–(d) Image displayed in 128, 64, and 32 gray levels, while keeping the spatial resolution constant.

Spatial and Gray level Resolution

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e f
g h

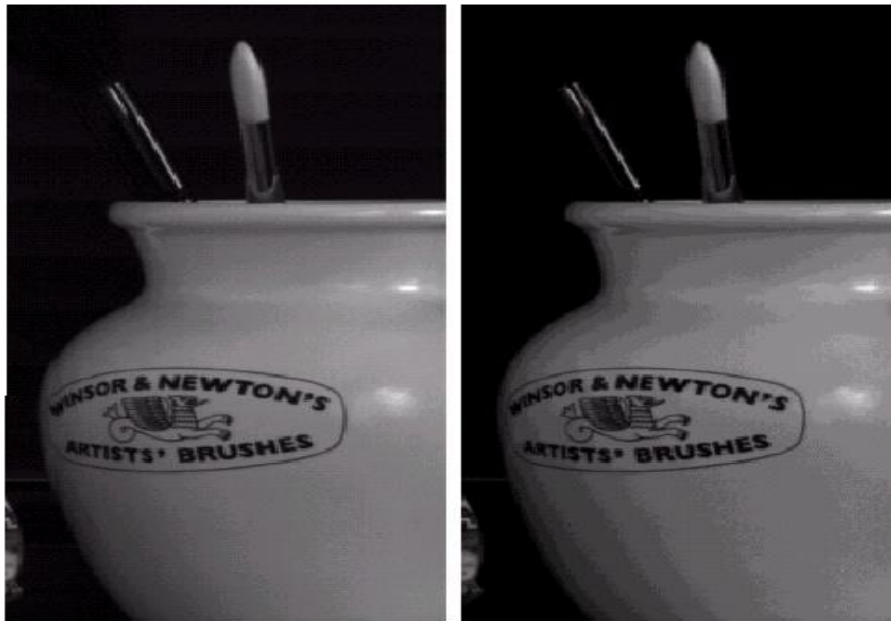
FIGURE 2.21
(Continued)
(e)–(h) Image displayed in 16, 8, 4, and 2 gray levels. (Original courtesy of Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)



Spatial and Gray level Resolution

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- ❑ The representation of an image with insufficient number of gray levels produces false edges or boundaries in an image, a phenomenon known as False Contouring or Contouring defect.
- ❑ False Contouring is quite visible in images displayed using 16 or less gray levels as shown in the images of the skeleton in the previous slide



Contouring
defect

Spatial and Gray level Resolution

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Medium level of detail

Low level of detail

Large amount of detail



a b c

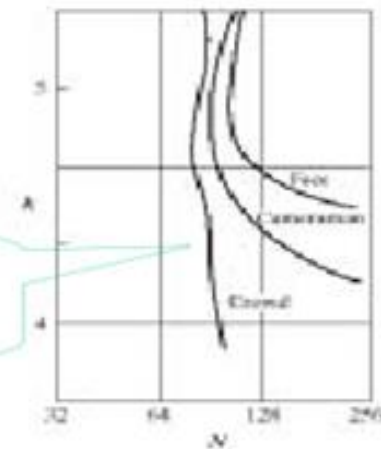
FIGURE 2.22 (a) Image with a low level of detail. (b) Image with a medium level of detail. (c) Image with a relatively large amount of detail. (Image (b) courtesy of the Massachusetts Institute of Technology.)

Spatial and Gray level Resolution

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- Varying N and k , the observers are then asked to rank the image according to their subjective quality.
- The results are summarized in the form of so-called **isopreference curve**.
- The **isopreference curve** tends to become more vertical as the detail of the image increases.

FIGURE 2.25
Representative
isopreference
curves for the
three types of
images in
Fig. 2.22.



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Aliasing and Moire Patterns

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- ❑ If the function is under-sampled, then a phenomena named Aliasing occurs
- ❑ The corruption is in the form of additional frequency components being introduced into the sampled function. These are called aliased frequencies
- ❑ The principal approach to reduce aliasing is to reduce the high frequency components prior to sampling
- ❑ Aliasing can be seen with Moire patterns



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IMAGE SCALING

Simple scaling



Image Scaling

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- ❑ Zooming (up scaling, resizing upward) requires two steps
 - ✓ The Creation of new pixel locations
 - ✓ Assignment of gray levels to new pixel locations

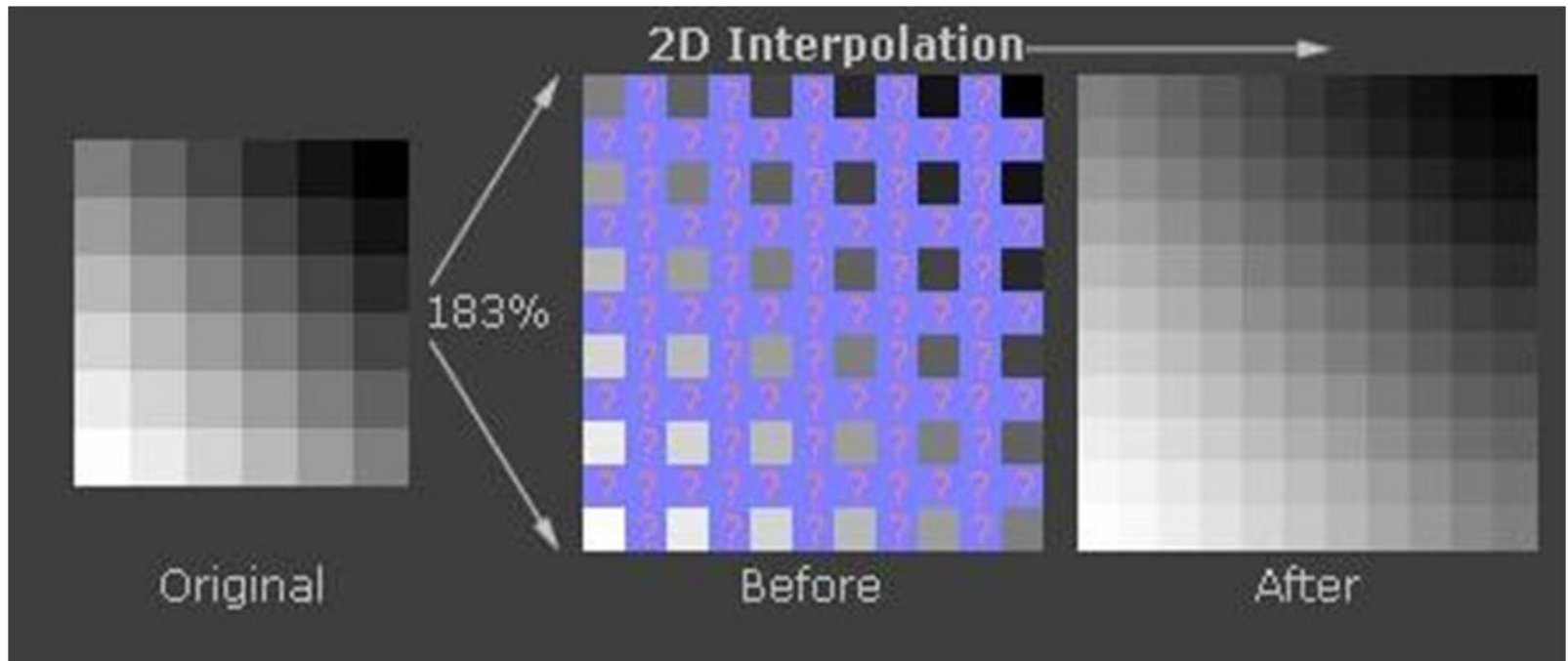


Image Scaling via Interpolation

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- ❑ Common interpolation algorithms can be grouped into two categories:
 - ✓ Adaptive and
 - ✓ Non-adaptive.

Adaptive vs Non-Adaptive Algorithm

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- ❑ **Adaptive algorithms** has the capability of applying different level of interpolation on different area on an image
- ❑ Adaptive methods change depending on what they are interpolating (sharp edges vs. smooth texture)
- ❑ It include many algorithms in licensed software such as: Qimage, PhotoZoom Pro, Genuine Fractals and others.
- ❑ **Non-adaptive algorithms** treat all pixels equally.
- ❑ The more adjacent pixels they include, the more accurate they can become, but this comes at the expense of much longer processing time.

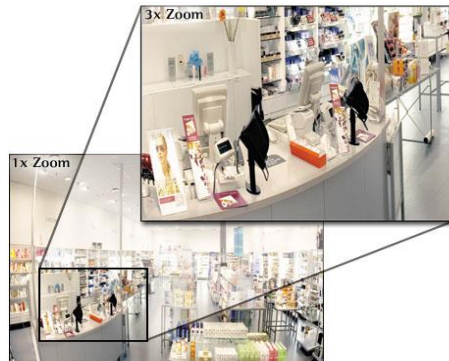


Image Scaling

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❑ Nearest neighbor Interpolation

- ✓ Nearest-neighbor interpolation (also known as proximal interpolation or, in some contexts, point sampling) is a simple method of multivariate interpolation in one or more dimensions.
- ✓ Interpolation is the problem of approximating the value of a function for a non-given point in some space when given the value of that function in points around (neighboring) that point. The nearest neighbor algorithm selects the value of the nearest point and does not consider the values of neighboring points at all
- ✓ Nearest neighbor is the simplest and fastest implementation of image scaling technique. It is very useful when speed is the main concern
- ✓ Nearest neighbor is the most basic and requires the least processing time of all the interpolation algorithms because it only considers one pixel

Image Scaling

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Pixel Replication

- ✓ Pixel replication is applicable when we want to increase the size of an image an integer number of times.
- ✓ For example, to double the size of an image we can duplicate each column, this doubles the size of image in horizontal direction. Then we duplicate each row of the enlarged image to double the size in the vertical direction
- ✓ The same procedure can be applied to enlarge the image by any integer number of times (triple, quadruple and so on)
- ✓ The gray level assignment of each pixel is predetermined by the fact that new locations are exact duplicate of old locations

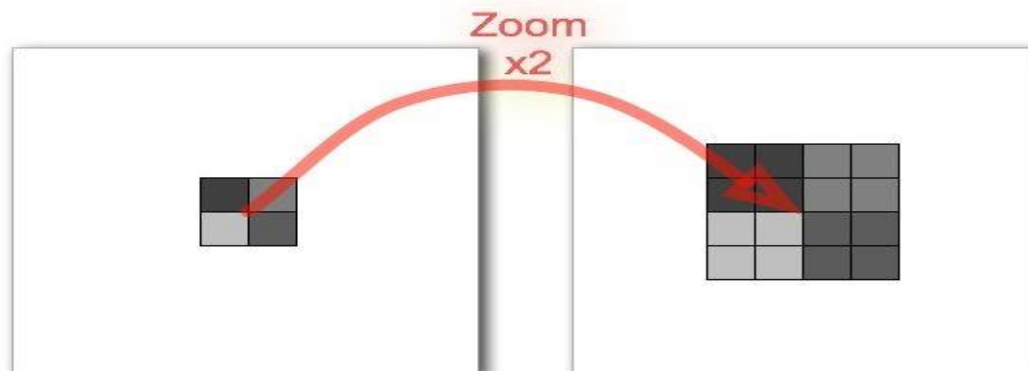


Image Scaling

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□ Bilinear Interpolation [2]

- Bilinear interpolation considers the closest 2x2 neighborhood of known pixel values surrounding the unknown pixel.
- It then takes a weighted average of these 4 pixels to arrive at its final interpolated value. This results in much smoother looking images than nearest neighbor.
- The diagram below is for a case when all known pixel distances are equal, so the interpolated value is simply their sum divided by four.
- In case the distance varies then the closer pixels are given more weightage in the calculation

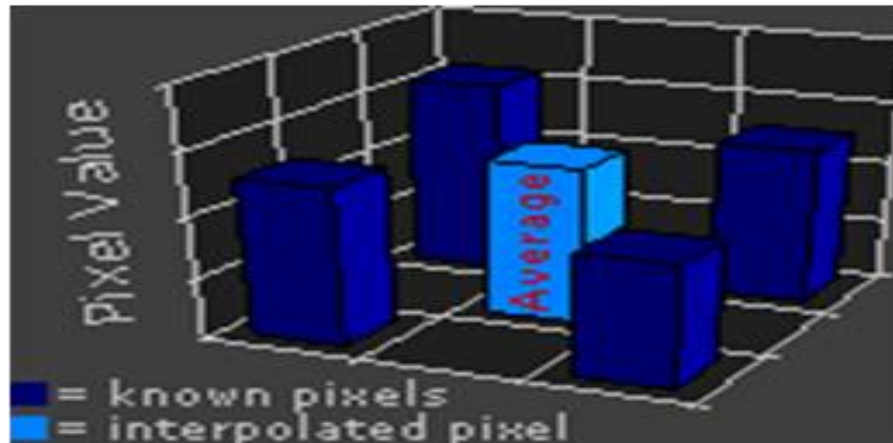


Image Scaling

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- The key idea is to perform linear interpolation first in one direction, and then again in the other direction.
- Suppose that we want to find the value of the unknown function f at the point $P = (x, y)$.
- It is assumed that we know the value of f at the four points $Q_{11} = (x_1, y_1)$, $Q_{12} = (x_1, y_2)$, $Q_{21} = (x_2, y_1)$, and $Q_{22} = (x_2, y_2)$.

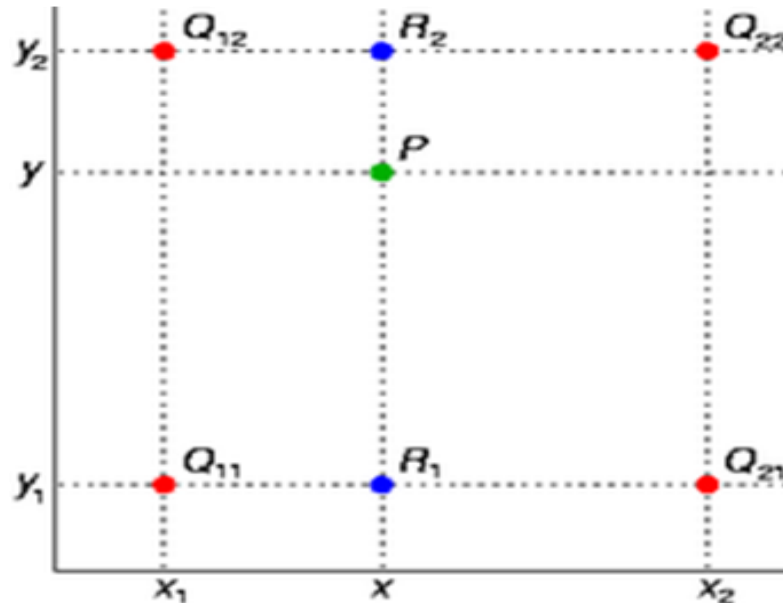
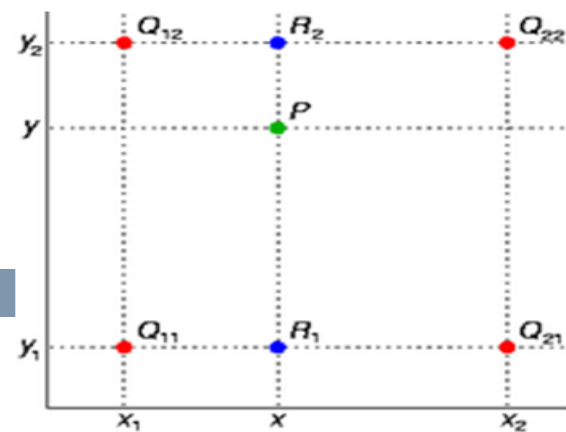


Image Scaling

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We first do linear interpolation in the x-direction. This yields

$$f(R_1) \approx \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21})$$

where $R_1 = (x, y_1)$,

$$f(R_2) \approx \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22})$$

where $R_2 = (x, y_2)$.

$$f(P) \approx \frac{y_2 - y}{y_2 - y_1} f(R_1) + \frac{y - y_1}{y_2 - y_1} f(R_2).$$

We proceed by interpolating in the y-direction.

Image Scaling

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❑ Bicubic Interpolation [3]

- ✓ Bicubic goes one step beyond bilinear by considering the closest 4x4 neighborhood of known pixels-- for a total of 16 pixels.
- ✓ Since these are at various distances from the unknown pixel, closer pixels are given a higher weighting in the calculation.
- ✓ Bicubic produces noticeably sharper images than the previous two methods, and is perhaps the ideal combination of processing time and output quality
- ✓ For this reason, it is a standard in many image editing programs (including Adobe Photoshop), printer drivers and in-camera interpolation

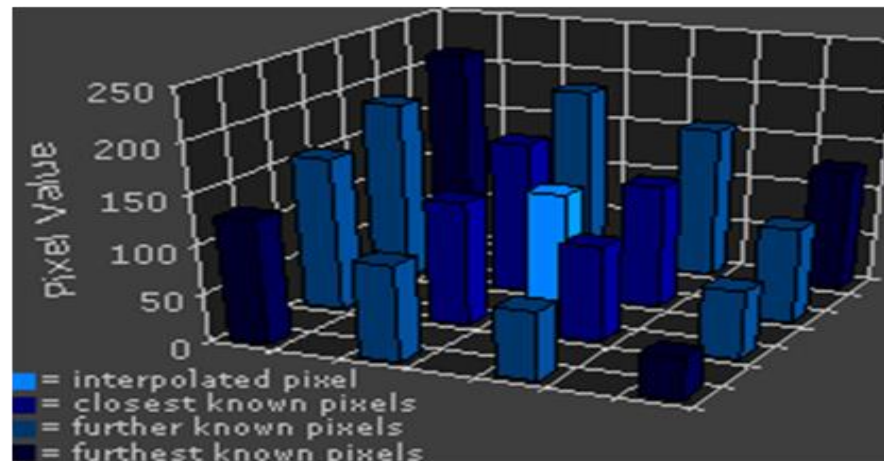
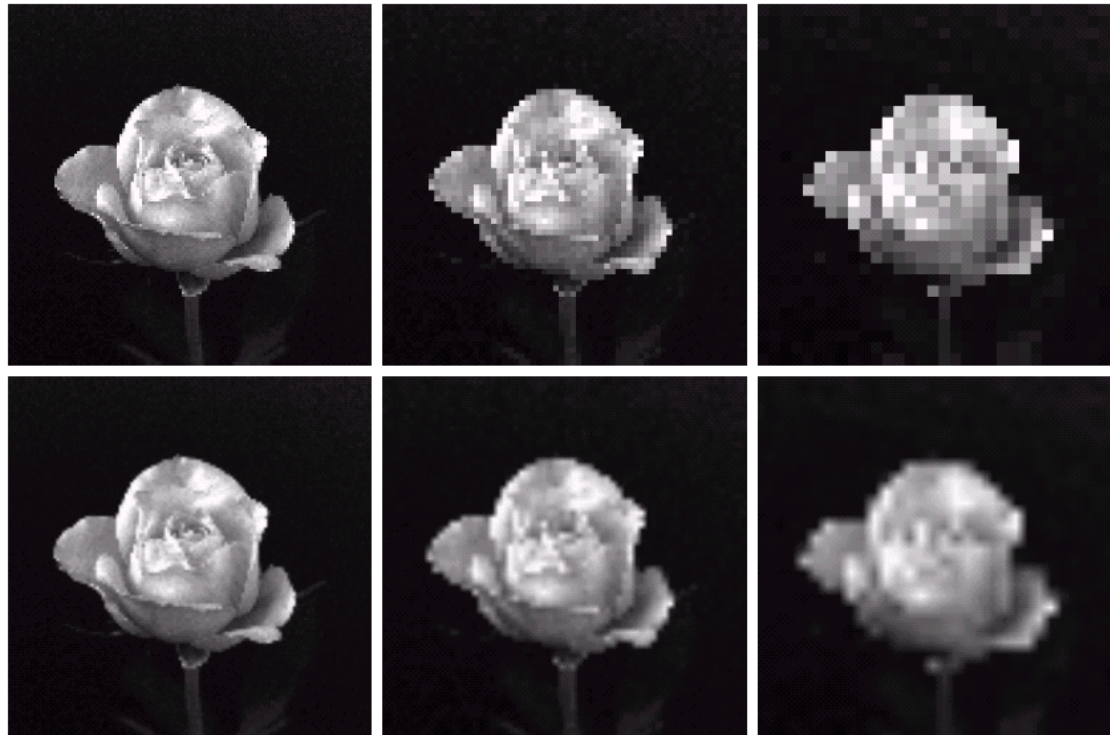


Image Scaling

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Increasing Spatial Resolution



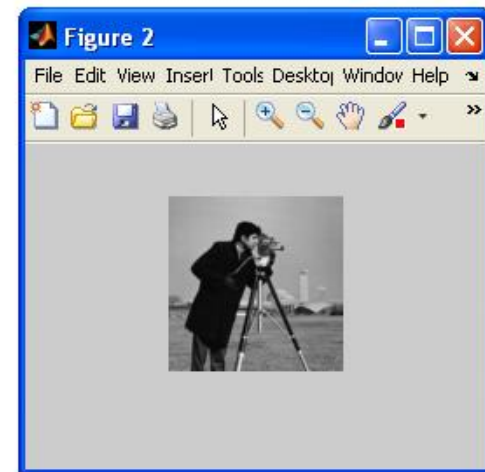
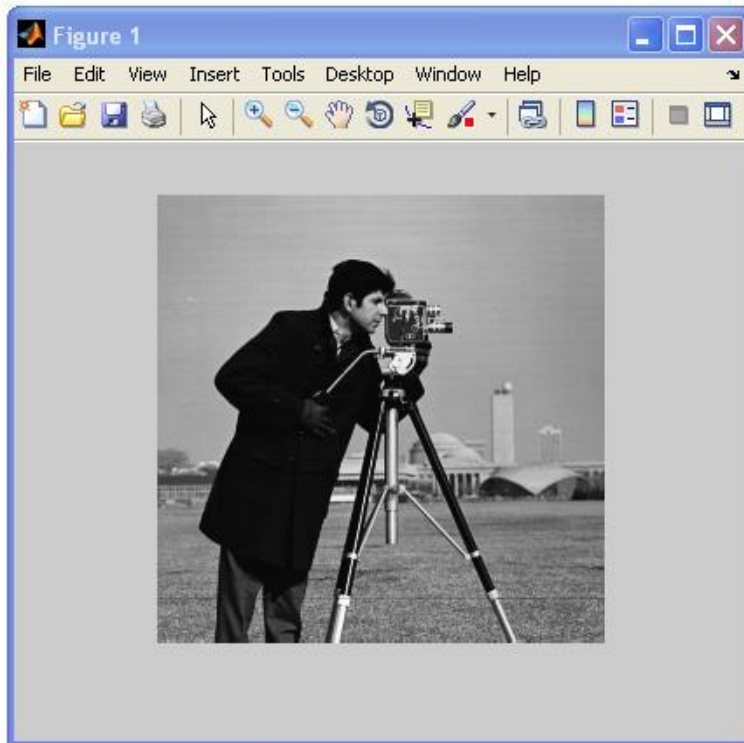
a	b	c
d	e	f

FIGURE 2.25 Top row: images zoomed from 128×128 , 64×64 , and 32×32 pixels to 1024×1024 pixels, using nearest neighbor gray-level interpolation. Bottom row: same sequence, but using bilinear interpolation.

Image Scaling

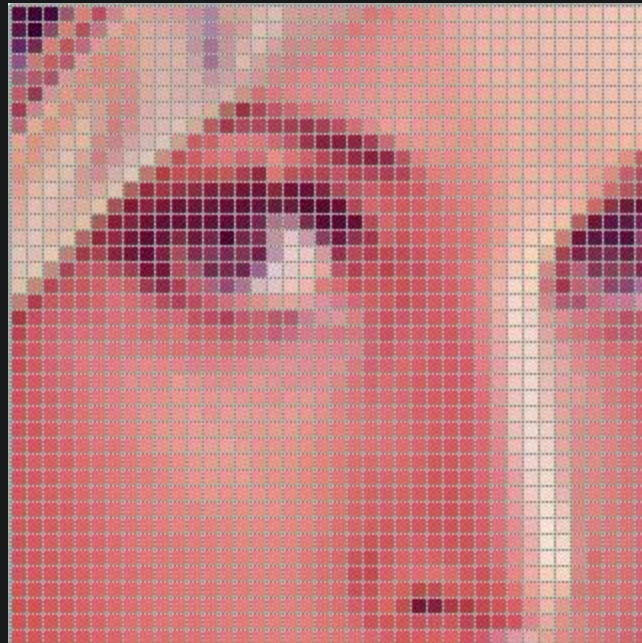
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- ❑ Shrinking (Down scaling, resizing downward)
 - ✓ Image shrinking is done in the similar manner as zooming with one difference as now the process of pixel replication is row column deletion. Now we can delete every second column and row for shrinking



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Basic Relationship between pixels



Basic Relationship between Pixels

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- ❑ Neighborhood
- ❑ Connectivity
- ❑ Adjacency
- ❑ Paths
- ❑ Regions and boundaries

0	1	0	1	1
1	1	1	0	1
0	1	0	1	1
0	1	1	0	1
0	0	1	0	0



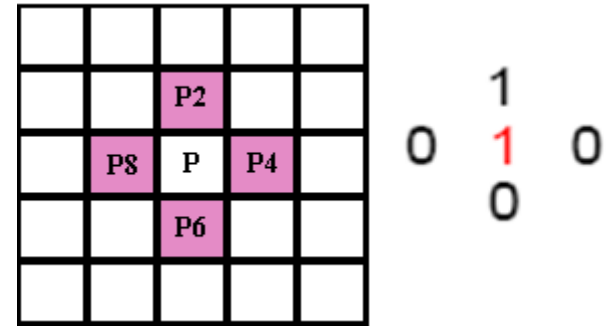
Pixels Neighborhood

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Neighbors of a pixel p at coordinates (x,y)

- ✓ 4-neighbors of p , denoted by $N_4(p)$:

$(x-1, y)$, $(x+1, y)$, $(x, y-1)$, and $(x, y+1)$



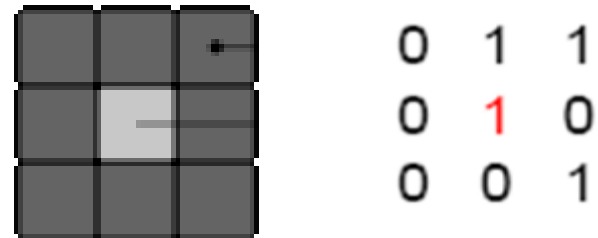
- ✓ 4 diagonal neighbors of p , denoted by $N_D(p)$:

$(x-1, y-1)$, $(x+1, y+1)$, $(x+1, y-1)$, and $(x-1, y+1)$



- ✓ 8 neighbors of p , denoted $N_8(p)$

$$N_8(p) = N_4(p) \cup N_D(p)$$



Connectivity & Adjacency

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- ❑ Two pixels are said to be connected if they are adjacent in some sense
 - ✓ They are neighbors
 - ✓ Their intensity values are the similar [gray levels are equal]
 - For example, in a binary image with values 0 and 1, two pixels may be 4-neighbors, but they are said to be connected only if they have the same value
- ❑ Let V be the set of gray level values used to define adjacency
- ❑ In a binary image, $V=\{1\}$ if we are referring to adjacency of pixels with value 1
- ❑ In gray scale image the idea is same but set V contains more elements. For example, in the adjacency of pixels with the range of possible gray level values 0 to 255, set V could be any subset of these 256 values

Connectivity & Adjacency

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- Let V be the set of intensity values
- **4-adjacency:** Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$
- **8-adjacency:** Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$
- **m-adjacency:** Two pixels p and q with values from V are m-adjacent if
 - (i) q is in the set $N_4(p)$, or
 - (ii) q is in the set $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V
 $q \in N_D(p)$ and $N_4(p) \cap N_4(q) = \emptyset$
- **Mixed adjacency** is a modification of 8-Adjacency. It is introduced to eliminate the ambiguities that often arise when 8-Adjacency is used

Path

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- A (digital) path (or curve) from pixel p with coordinates (x, y) to pixel q with coordinates (s, t) is a sequence of distinct pixels with coordinates $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$

Where $(x_0, y_0) = (x, y)$, $(x_n, y_n) = (s, t)$

(x_i, y_i) and (x_{i-1}, y_{i-1}) are adjacent for $1 \leq i \leq n$.

- Here n is the *length* of the path.
- If $(x_0, y_0) = (x_n, y_n)$, the path is *closed* path.
- We can define 4-, 8-, and m-paths based on the type of adjacency used.

Connectivity & Adjacency

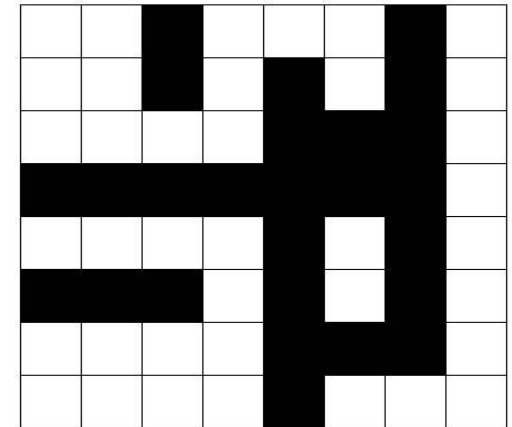
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- Let S represents a subset of pixels in an image. Two pixels p with coordinates (x_0, y_0) and q with coordinates (x_n, y_n) are said to be **connected in S** if there exists a path between them consisting entirely of pixels in S

e.g $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$

where

$$\forall i, 0 \leq i \leq n, (x_i, y_i) \in S$$



- For any pixel p in S , the set of pixels that are connected to it in S is called a **connected component** of S
- If it only has one connected component, then set S is called **Connected set**

Examples: Adjacency and Path

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$$V = \{1, 2\}$$

0 1 1

0 1 1

0 1 1

0 2 0

0 2 0

0 2 0

0 0 1

0 0 1

0 0 1

Examples: Adjacency and Path

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$$V = \{1, 2\}$$

0 1 1

0 2 0

0 0 1

0 1 1

0 2 0

0 0 1

8-adjacent

0 1 1

0 2 0

0 0 1

Examples: Adjacency and Path

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$$V = \{1, 2\}$$

0	1	1
0	2	0
0	0	1

0	1	1
0	2	0
0	0	1

8-adjacent

0	1	1
0	2	0
0	0	1

m-adjacent

Examples: Adjacency and Path

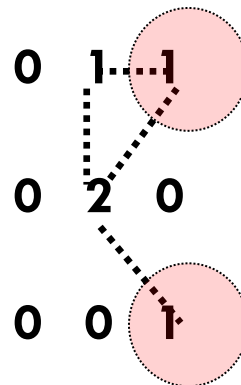
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$$V = \{1, 2\}$$

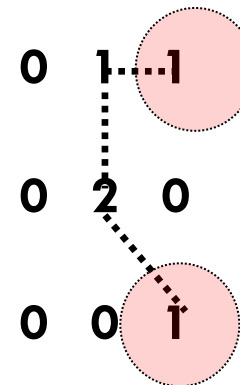
0 1 1

0 2 0

0 0 1



8-adjacent



m-adjacent

The 8-path from (1,3) to (3,3):

(i) (1,3), (1,2), (2,2), (3,3)

(ii) (1,3), (2,2), (3,3)

The m-path from (1,3) to (3,3):

(1,3), (1,2), (2,2), (3,3)

Connected Component Labeling

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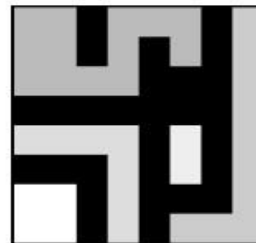
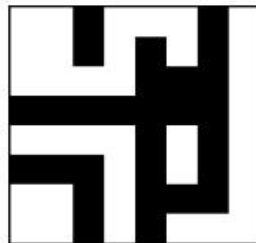
- Ability to assign different labels to various disjoint connected components of an image
- Connected component labeling is a fundamental step in automated image analysis (Shape, Area, Boundary)

1	1	0	1	1	1	0	1
1	1	0	1	0	1	0	1
1	1	1	1	0	0	0	1
0	0	0	0	0	0	0	1
1	1	1	1	0	1	0	1
0	0	0	1	0	1	0	1
1	1	0	1	0	0	0	1
1	1	0	1	0	1	1	1

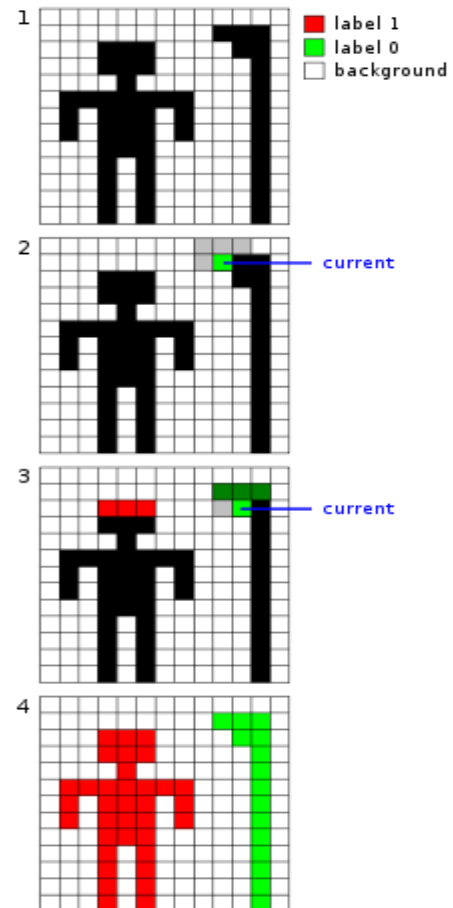
a) binary image

1	1	0	1	1	1	0	2
1	1	0	1	0	1	0	2
1	1	1	1	0	0	0	2
0	0	0	0	0	0	0	2
3	3	3	3	0	4	0	2
0	0	0	3	0	4	0	2
5	5	0	3	0	0	0	2
5	5	0	3	0	2	2	2

b) connected components labeling



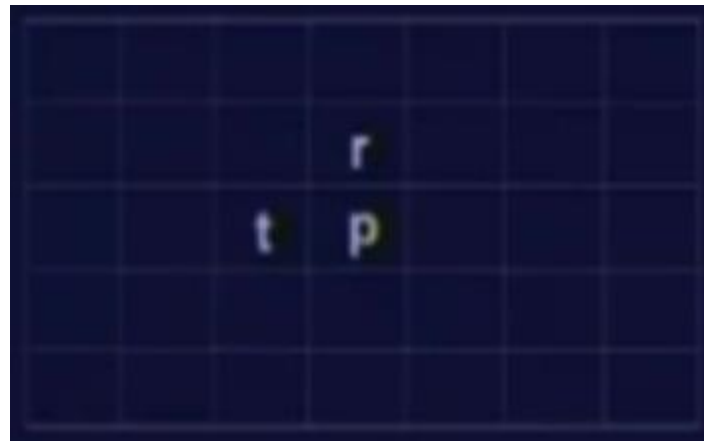
c) binary image and labeling, expanded for viewing



Connected Component Labeling

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- ❑ Scan an image from left to right and from top to bottom.
- ❑ Assume 4-Connectivity
- ❑ P be a pixel of any step in the scanning process



- ❑ Before p , points r and t are scanned.

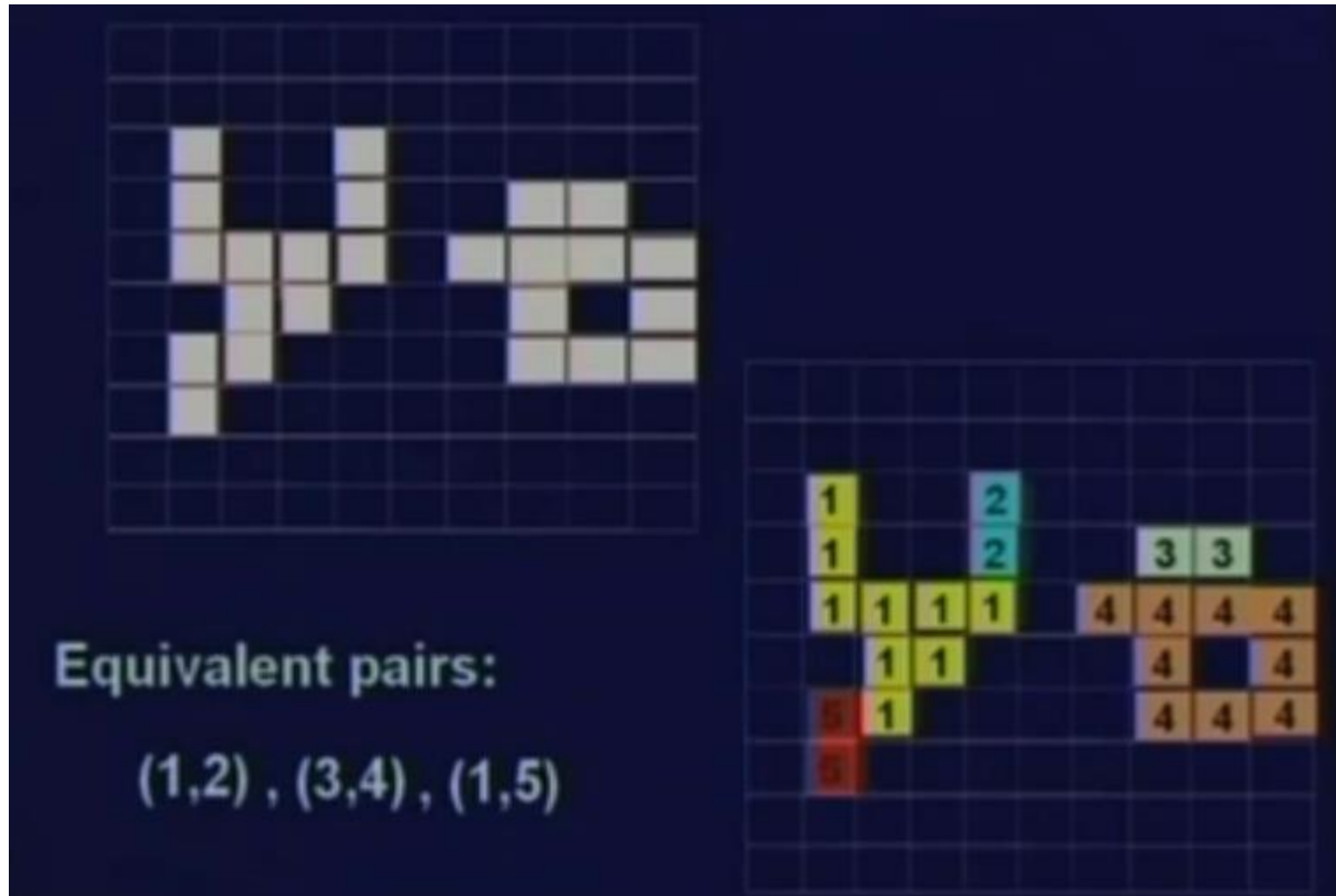
Connected Component Labeling

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- ❑ $I(p) \Rightarrow$ pixel value at position p .
- ❑ $L(p) \Rightarrow$ label assigned to pixel location p .
- ❑ If $I(p) = 0$, move to next scanning position.
- ❑ If $I(p) = 1$ and $I(r) = I(t) = 0$
 - ✓ Then assign a new label to position p .
- ❑ If $I(p) = 1$ and only one of the two neighbor is 1 .
 - ✓ Then assign its label to p .
- ❑ If $I(p) = 1$ and both r and t are 1's, then
- ❑ If $L(r) = L(t)$ then $L(p) = L(r) = L(t)$ where $[L(r)=L(t)]$
- ❑ If $L(r) \neq L(t)$ then assign one of the labels to p and make a note that the two labels are equivalent.

Connected Component Labeling

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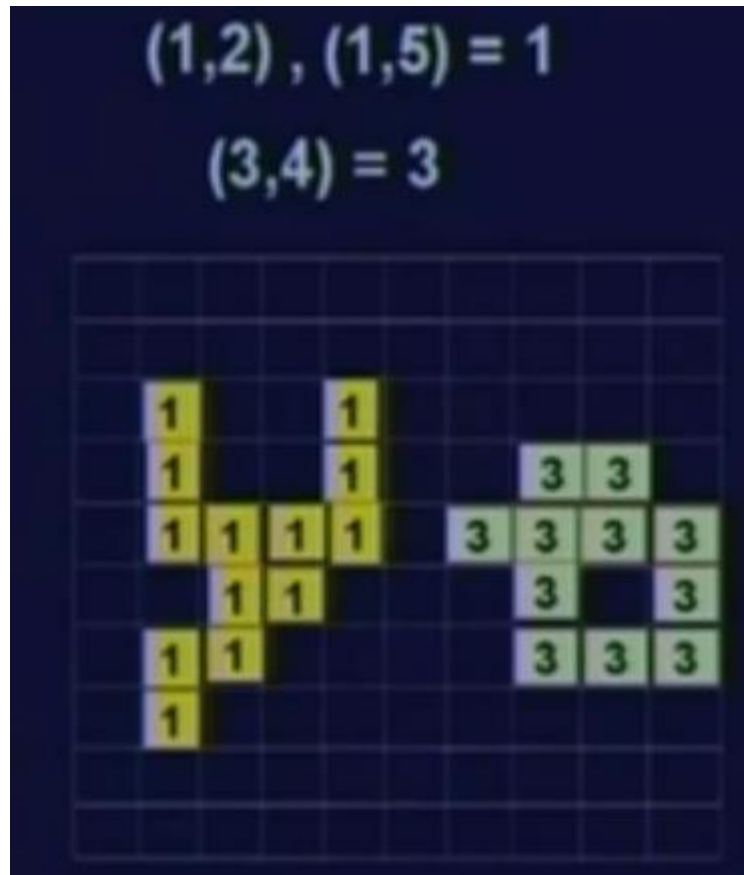
Connected Component Labeling

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- ❑ At end of the scan all pixels with value 1 are labeled
- ❑ Some labels are equivalent.
- ❑ During second pass process equivalent pairs to form equivalence classes.
- ❑ Assign a different label to each class
- ❑ In the second pass through the image replace each label by the label assigned to its equivalence classes.

Connected Component Labeling

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Regions & Boundaries

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- ❑ Let R represent a subset of pixels in an image
- ❑ We call R a region of the image if R is a connected set
- ❑ We can specify the region by using 4-adjacency and 8-adjacency
- ❑ Region = { Set of all pixels which fulfill adjacency criteria }
- ❑ Boundary (or border)
 - ✓ The *boundary* of the region R is the set of pixels in the region that have one or more neighbors that are not in R .
 - ✓ If R happens to be an entire image, then its boundary is defined as the set of pixels in the first and last rows and columns of the image.
- ❑ Foreground and background
 - ✓ An image contains K disjoint regions, $R_k, k = 1, 2, \dots, K$. Let R_u denote the union of all the K regions, and let $(R_u)^c$ denote its complement
 - ✓ All the points in R_u is called foreground;
 - ✓ All the points in $(R_u)^c$ is called background.

Distance measures

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- For pixels p , q and z with coordinates (x, y) , (s, t) , and (v, w) , respectively, D is a distance function or metric if
 - ✓ $D(p, q) \geq 0$ ($D(p, q) = 0$ if $p=q$)
 - ✓ $D(p, q) = D(q, p)$
 - ✓ $D(p, z) \leq (D(p, q) + D(q, z))$
- It is often useful in image processing to be able to calculate the distance between two pixels in an image, but this is not as straightforward as it seems.
- The presence of the pixel grid makes several so-called *distance metrics* possible which often give different answers to each other for the distance between the same pair of points. We consider the three most important ones.
 - ✓ Euclidean distance
 - ✓ City-block distance or D_4 distance.
 - ✓ D_8 distance or chessboard distance.

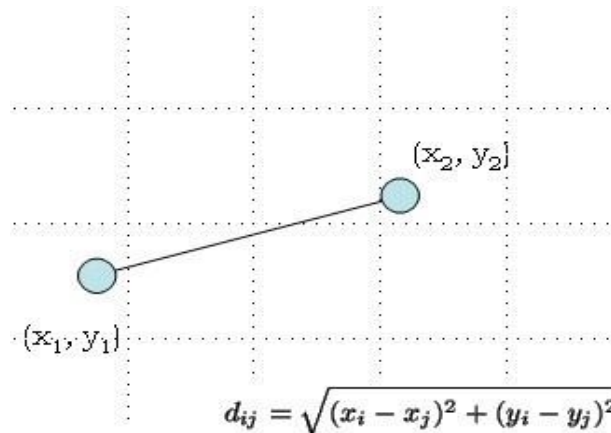
Distance measures

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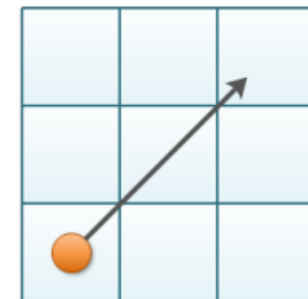
□ Euclidean distance

- ✓ If the two pixels p and q that we are considering have coordinates (x,y) and (s,t) , then the Euclidean distance between p and q is given by:

$$D_e(p, q) = [(x-s)^2 + (y-t)^2]^{1/2}$$



Euclidean Distance



$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Distance measures

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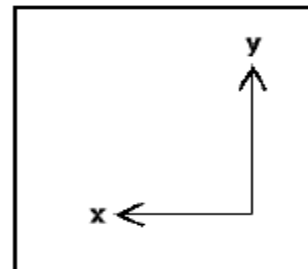
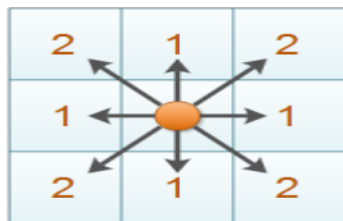
❑ City-block distance or D4 distance or Manhattan distance

- ✓ This metric assumes that in going from one pixel to the other it is only possible to travel directly along pixel grid lines. Diagonal moves are not allowed.
- ✓ The distance between two points is the sum of the absolute differences of their coordinates
- ✓ The City-block distance between p and q is defined as

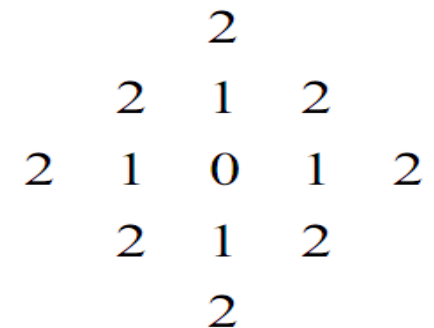
$$D4(p, q) = |x - s| + |y - t|$$

- ✓ The pixels having a distance $D4$ from (x, y) less than or equal to some value r form a diamond centered at (x, y) . E.g the pixels with $D4$ distance ≤ 2 from (x, y) (the center point) form the following contours of constant distance

Manhattan Distance



Manhattan



Distance measures

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❑ D8 distance or chessboard distance or Chebyshev distance

- ✓ This metric assumes that you can make moves on the pixel grid as if you were a King making moves in chess, i.e., a diagonal move counts the same as a horizontal move
- ✓ The distance between two pixels is the greatest of their differences of their coordinates

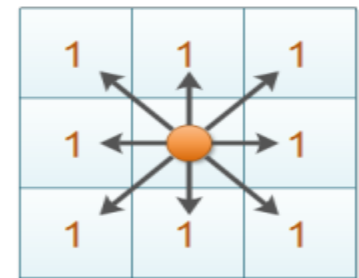
$$D8(p, q) = \max(|x - s|, |y - t|)$$

- ✓ The pixels having a D8 distance from (x, y) less than or equal to some value r form a square centered at (x, y) . E.g the pixels with D8 distance ≤ 2 from (x, y) (the center point) form the following contours of constant distance

- ✓ The pixels with $D8 = 1$ are the 8-neighbors of (x, y)

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Chebyshev Distance

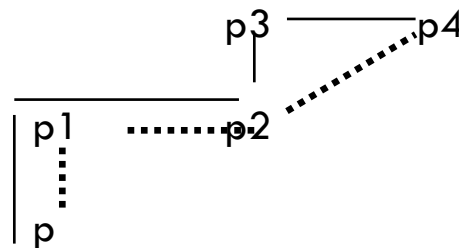


$$\max(|x_1 - x_2|, |y_1 - y_2|)$$

Distance measures and Connectivity

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- Consider the following arrangement of pixels and assume that p, p_2 and p_4 have value 1 and p_1 and p_3 have values 0 or 1:

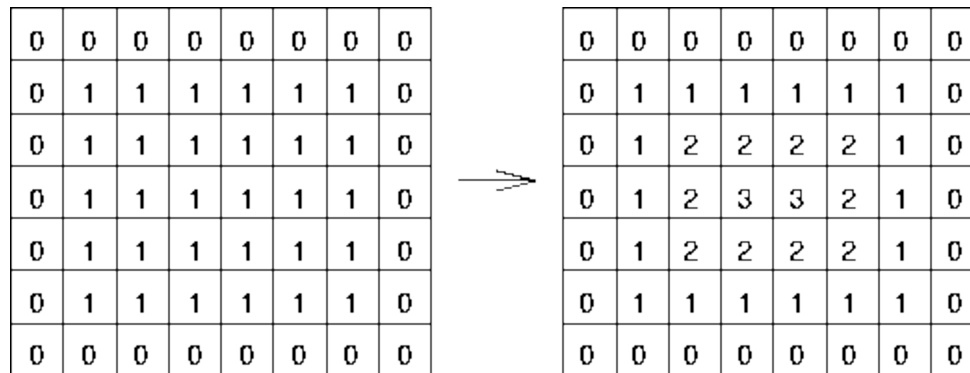


- Suppose we have $V=\{1\}$ [Adjacency criteria]
- If p_1 and p_3 are zero, the length of the shortest m-path (D_m distance) between p and p_4 is _____
- If $p_1=1$ and $p_3=0$, then p_2 and p will no longer be m-adjacent, and the length of the shortest m-path becomes _____ [Path will be $p \ p_1 \ p_2 \ p_4$]
- If $p_1=0$ and $p_3=1$, then the length of the shortest m-path will be _____
- Finally, if both p_1 and p_3 are 1 then the length of the shortest m-path between p and p_4 is _____ [Path will be $p \ p_1 \ p_2 \ p_3 \ p_4$]

Distance Transform [6]

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- The distance transform is an operator normally only applied to binary images.
- The result of the transform is a gray level image that looks similar to the input image, except that the gray level intensities of points inside foreground regions are changed to show the distance to the closest boundary from each point.



The distance transform of a simple shape by using the chessboard distance metric.

Image Operation on Pixel basis

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- ❑ As we know images are represented in the form of matrices with Rows and Columns arrangements
- ❑ Any operation either arithmetic or logical is carried out between corresponding pixels in the images
- ❑ For example, if you want to add two images a and b then the resultant image will be formed in a way that the first element will be formed by the addition of the first pixel in a with the first pixel in b and so on for each pixel
- ❑ Similarly, subtraction, multiplication and division are carried out in a similar manner
- ❑ The logical operators AND, OR and NOT are applied on only one image at a time

References

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5. <http://www.mathworks.com/help/toolbox/images/f18-34186.html>
6. <http://homepages.inf.ed.ac.uk/rbf/HIPR2/distance.htm>

For any query Feel Free to ask



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